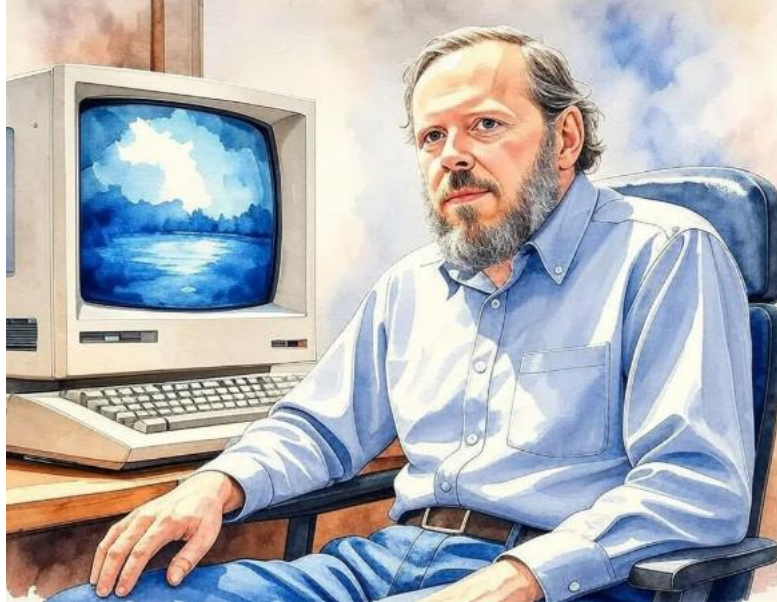


## CS25C01- Computer Programming: C

### Unit- I INTRODUCTION TO C



The C programming language is a structure oriented programming language developed at AT & T's Bell Laboratories of USA in 1972

It was designed and written by a man named **Dennis Ritchie**

#### Applications of C programming

- **Operating Systems:** Linux kernel, Microsoft Windows, and Apple macOS/iOS.
- **Databases:** MySQL, PostgreSQL, and Oracle
- **Compilers and Interpreters:** The standard Python interpreter (**CPython**) is written in C. CPython is the actual program that executes your Python code.
- **Embedded Systems:** Microwaves, washing machines, and digital watches.
- **IoT (Internet of Things):** Smart devices, sensors, and home automation systems.
- **Device Drivers:** Graphics cards, keyboards, and printers
- **Game Development:**
- **Graphics and Animation:** Frameworks like **OpenGL** and **DirectX**, which render 2D and 3D graphics for movies and software, are built using C concepts.
- **Web Browsers:** **Google Chrome** and **Mozilla Firefox**

**Problem Solving** means converting a real-world problem into a step-by-step solution that a computer can understand and execute.

The problem solving process is a systematic way to convert a problem into a working program.

## Steps for Problem Solving

### Problem Definition (Understand the Problem)

Read the question carefully

Identify:

- **Input**
- **Output**

Clarify what exactly is required

### Problem Analysis (Analysing the Problem)

Think about **how to solve it**

Break the problem into smaller parts

Decide:

- What logic is needed?
- Any formulas, conditions, or loops?

### Algorithm Design (Pseudo code)

Write step-by-step instructions in simple English

Example:

Start

Read input

Process data

Display result

Stop

### Flowchart

Draw diagram using symbols:

- Oval → Start/Stop
- Rectangle → Process
- Diamond → Decision

Helps visualize logic clearly

### Coding (Implementation)

- Convert the algorithm into a program (e.g., C language)

### Testing & Debugging

- Run program with different inputs
- Check:
  - Correct output?
  - Any errors?

### Documentation & Maintenance

- Add comments
- Improve code if needed
- Make the code readable

## Real-Life Example: Making Tea

### Understanding the Problem

Prepare a cup of tea

- Input → Water, milk, tea powder, sugar
- Output → Ready tea

### Analysing the Problem

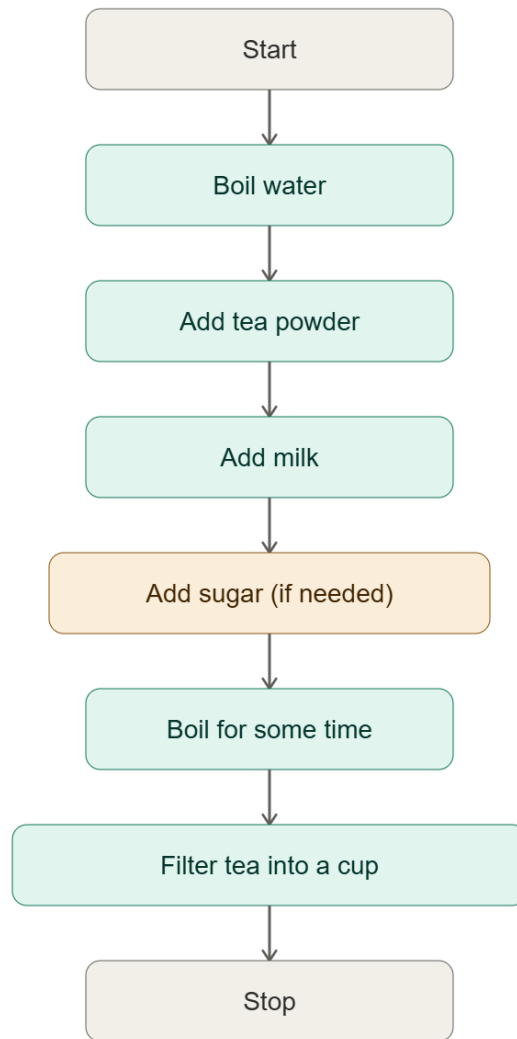
Think:

- What steps are needed?
- In what order?
- Any conditions? (e.g., sugar optional)

### Designing the Solution (Algorithm)

1. Start
2. Boil water
3. Add tea powder
4. Add milk
5. Add sugar (if needed)
6. Boil for some time
7. Filter tea into a cup
8. Stop

### Flowchart



### Coding (Implementation)

Prepare the tea as per the flowchart

### Testing & Debugging

- Taste the tea
- Too strong → reduce tea powder
- Less sweet → add sugar

This is like fixing errors in a program!

### Maintenance

- Next time:
  - Improve taste
  - Adjust ingredients

**Problem: Find the Square of a Number**

### Understanding the Problem

- What is required? → Find the **square of a number**
- Input → A number  $n$
- Output →  $n \times n$  (square)

Example:

If  $n = 5$ , output = 25

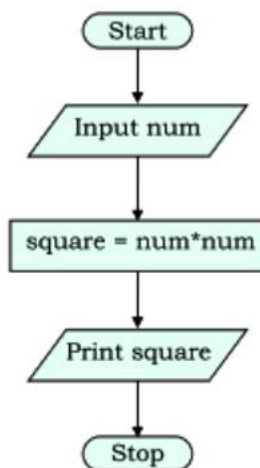
### Analyzing the Problem

- We need:
  - One variable  $\rightarrow n$
  - One operation  $\rightarrow$  multiplication

### Designing the Algorithm

1. Start
2. Read the number  $n$
3. Compute square =  $n \times n$
4. Display square
5. Stop

### Flowchart



### Coding (Implementation)

```
#include <stdio.h>
int main() {
    int n, square;
    printf("Enter a number: ");
    scanf("%d", &n);
    square = n * n;
    printf("Square = %d", square);
    return 0;
}
```

### Testing & Debugging

#### Test Case 1:

Input: 4

Output: 16

#### Test Case 2:

Input: -3

Output: 9

### Maintenance

- Improve program:
  - Handle large numbers
  - Add comments
  - Use meaningful variable names



### Find the Area of a Circle

**Problem:** Calculate the area of a circle.

- **Input:** Radius  $r$
- **Process:**  $\text{area} = 3.14 \times r \times r$
- **Output:** Area of the circle

**Example:**

Input: 5

Output: 78.5